Unit BMM350

SKU:U231























Description

Unit BMM350 is a three-axis magnetometer unit featuring the BMM350 sensor and supporting 100 kHz and 400 kHz communication rates. Based on TMR (tunnel magnetoresistance) technology, it measures magnetic fields along the X/Y/Z axes with approximately 0.1 µT resolution and excellent temperature stability. The module transmits data over I2C, provides real-time three-axis geomagnetic data, and supports heading detection after calibration, making it suitable for robot navigation, heading detection, electronic compass, motion tracking, and magnetic field sensing.

Features

- BMM350 three-axis TMR magnetometer
- Measurement axes: X / Y / Z
- Magnetic field resolution: approx. 0.1 µT
- Magnetic field range: ±2000 µT (typical)
- Magnetic sensing technology: TMR (tunnel magnetoresistance)
- Communication rate: 100 kHz / 400 kHz
- Supports heading readout and calibration

Includes

- 1 x Unit BMM350
- 1 x HY2.0-4P Grove Cable (5cm)

Applications

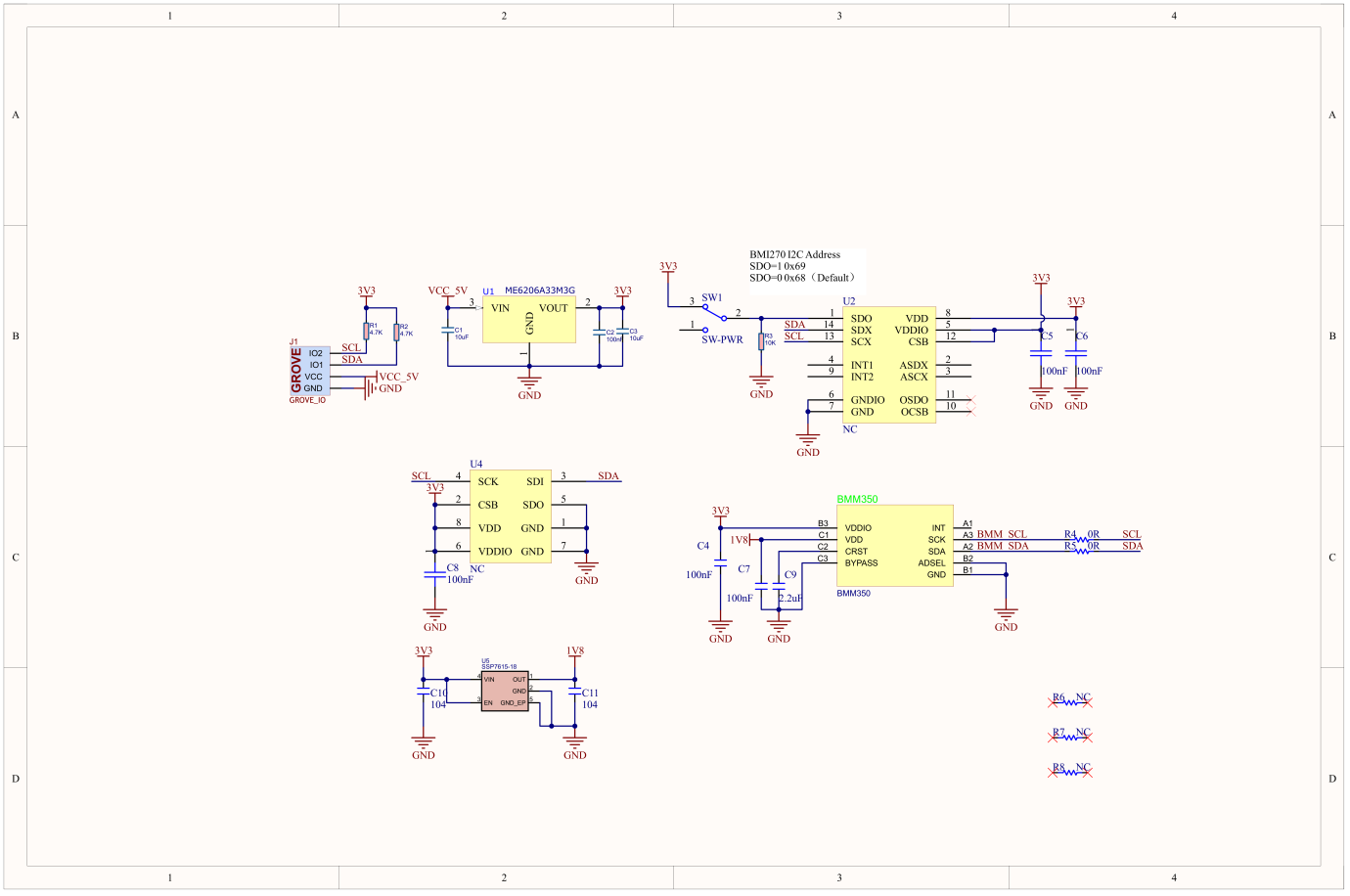
- Robot navigation
- Heading detection
- Electronic compass
- Motion tracking
- Magnetic field sensing

Specifications

Specification	Parameter
IC Model	BMM350
Sensor Type	Three-axis magnetometer
Magnetic Field Range	X/Y/Z axes: ±2000 μT (typical on each axis)
Magnetic Field Resolution	approx. 0.1 μT
Communication Interface	I2C @ 0x14
I2C Communication Rate	100 kHz / 400 kHz
Sleep Current	6.42 μA (measured)
Continuous Read Current	167.02 μA (measured)
Heading Accuracy	Error < 10° after calibration
Operating Temperature Range	-40 ~ 85°C (chip specification)
Product Size	24.0 x 24.0 x 8.0mm
Product Weight	3.1g
Package Size	138.0 x 93.0 x 7.0mm
Gross Weight	6.6g

Schematics

- Unit BMM350 Schematics PDF



PinMap

Unit BMM350

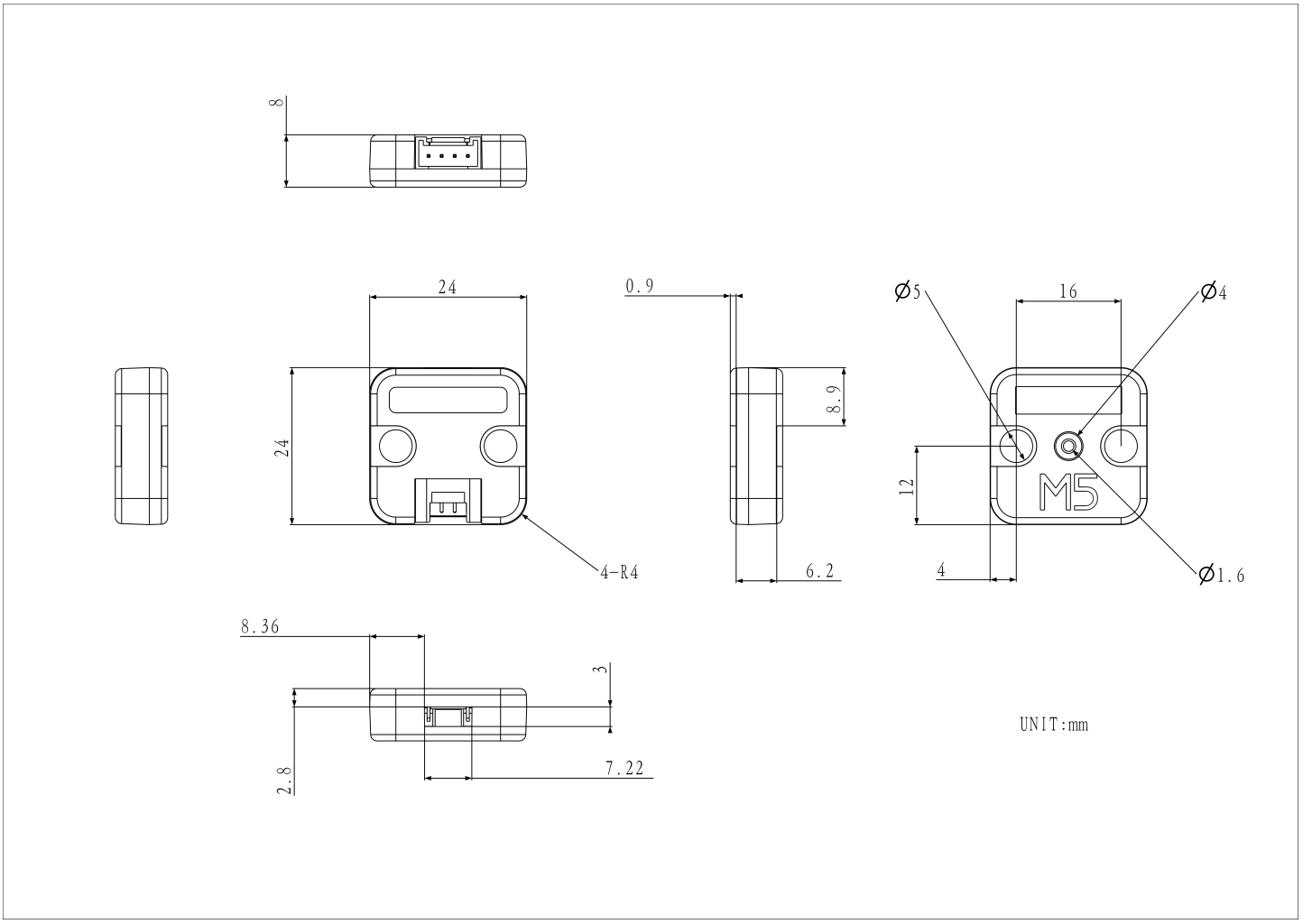
HY2.0-4P	Black	Red	Yellow	White
PORT.A	GND	5V	SDA	SCL

Datasheets

- BMM350 Datasheet

Model Size

- BMM350 Model Size PDF



Structure

- Unit BMM350 Structure Files

Softwares

Arduino

- Unit BMM350 Arduino Quick Start
- Unit BMM350 Arduino Library
- M5Unified
- M5UnitUnified

UiFlow2

- Unit BMM350 UiFlow2 Docs

Video

Product Comparison

Product Comparison				
	Unit DoF6	**Unit DoF9**	**Unit DoF10**	**Unit BMM350**
Sensor Combination	BMI270	BMI270 + BMM350	BMI270 + BMM350 + SPA06-003	BMM350
Sensor Type	Six-axis inertial sensor	Six-axis inertial + three-axis geomagnetic sensor	Six-axis inertial + three-axis geomagnetic + barometric pressure sensor	Three-axis geomagnetic sensor
Accelerometer	16-bit 3-axis	16-bit 3-axis	16-bit 3-axis	❌
Gyroscope	16-bit 3-axis	16-bit 3-axis	16-bit 3-axis	❌
Magnetometer	❌	BMM350 (TMR, approx. 0.1 μT)	BMM350 (TMR, approx. 0.1 μT)	BMM350 (TMR, approx. 0.1 μT)
Barometric Pressure Sensor	❌	❌	SPA06-003	❌
I2C Address	0x68 / 0x69 (onboard switch)	BMI270: 0x68 / 0x69 (onboard switch) BMM350: 0x14	BMI270: 0x68 / 0x69 (onboard switch) BMM350: 0x14 SPA06-003: 0x76	0x14
Main Functions	Acceleration, angular velocity, and temperature acquisition; on-chip motion detection	Acceleration, angular velocity, temperature, and three-axis magnetic field acquisition; on-chip motion detection	Acceleration, angular velocity, temperature, three-axis magnetic field, and pressure acquisition; on-chip motion detection	Three-axis magnetic field acquisition; heading output after calibration